

Categories, Proofs, and Processes:

Assignment 4

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Question 1

Proposition 1.1. *The covariant Hom-functor $\mathcal{C}(A, -) : \mathcal{C} \rightarrow \mathbf{Set}$ preserves all finite limits.*

Proof. Let $D : \mathcal{J} \rightarrow \mathcal{C}$ be a diagram where $\mathcal{J}$ is a finite category. Suppose the cone $(C, (c_1, c_2, \dots, c_N))$ to D is a finite limit. By being a cone, for any $f : I \rightarrow J$ in $\mathcal{J}$, we have

$$\begin{array}{ccc} D I & \xrightarrow{D f} & D J \\ & \swarrow c_I \quad \searrow c_J & \\ & C & \end{array}$$

$$D f \circ c_I = c_J$$

We show that $(\mathcal{C}(A, C), (\mathcal{C}(A, c_1), \dots, \mathcal{C}(A, c_N)))$ is a cone and a limit to $\mathcal{C}(A, -) \circ D : \mathcal{J} \rightarrow \mathbf{Set}$.

$$\mathcal{C} \xrightarrow{\mathcal{C}(A, -)} \mathbf{Set}$$

$$\begin{array}{ccc} C & \xrightarrow{\quad} & \mathcal{C}(A, C) \\ c_I \downarrow & \xrightarrow{\quad} & \downarrow \mathcal{C}(A, c_I) \\ D I & \xrightarrow{\quad} & \mathcal{C}(A, D I) \\ D f \downarrow & \xrightarrow{\quad} & \downarrow \mathcal{C}(A, D f) \\ D J & \xrightarrow{\quad} & \mathcal{C}(A, D J) \end{array}$$

Suppose we have an arbitrary $f : I \rightarrow J$ in $\mathcal{J}$, so we get

$$\mathcal{C}(A, (D f)) : \mathcal{C}(A, (D I)) \rightarrow \mathcal{C}(A, (D J))$$

Consider the morphisms $D I \xleftarrow{c_I} C \xrightarrow{c_J} D J$, so

$$\mathcal{C}(A, c_I) : \mathcal{C}(A, C) \rightarrow \mathcal{C}(A, (D I)) := h \mapsto c_I \circ h$$

$$\mathcal{C}(A, c_J) : \mathcal{C}(A, C) \rightarrow \mathcal{C}(A, (D J)) := h' \mapsto c_J \circ h'$$

$$\begin{array}{ccc}
\mathcal{C}(A, D I) & \xrightarrow{\mathcal{C}(A, D f)} & \mathcal{C}(A, D J) \\
& \nwarrow \mathcal{C}(A, c_I) \quad \nearrow \mathcal{C}(A, c_J) & \\
& \mathcal{C}(A, C) &
\end{array}$$

Then the diagram commutes since

$$\begin{aligned}
\mathcal{C}(A, (D f)) \circ \mathcal{C}(A, c_I) &= (h \mapsto (D f) \circ h) \circ (h' \mapsto c_I \circ h') \\
&= h \mapsto (D f) \circ c_I \circ h \\
&= h \mapsto c_J \circ h \\
&= \mathcal{C}(A, c_J)
\end{aligned} \tag{1}$$

So $(\mathcal{C}(A, C), (\mathcal{C}(A, c_1), \dots, \mathcal{C}(A, c_N)))$ is a cone to $\mathcal{C}(A, -) \circ D$.

To show the limit, consider $\mathcal{C}(A, C)$ in **Set** and let (B, γ) be some other cone to $\mathcal{C}(A, -) \circ D$, where γ is

$$\{\gamma_I : B \rightarrow \mathcal{C}(A, (D I))\}_{I \in \text{Ob}(\mathcal{J})}$$

Since $(C, (c_1, c_2, \dots, c_N))$ is a limit to D , for any cone (A, δ) to D , there is a unique arrow $a : A \rightarrow C$ such that $\forall I \in \text{Ob}(\mathcal{J}), c_I \circ a = \delta_I$.

Note that $\forall b \in B, \gamma_I(b) \in \mathcal{C}(A, (D I))$. Then for each $b \in B$, we have the cone

$$(A, \{\gamma_I(b) : A \rightarrow D I\}_{I \in \text{Ob}(\mathcal{J})})$$

From being the limit, there is a unique map $u_b : A \rightarrow C$ such that

$$\begin{array}{ccc}
& & D I \\
& \nearrow \gamma_I(b) & \uparrow c_I \\
A & \xrightarrow{u_b} & C
\end{array}$$

$$\forall I \in \text{Ob}(\mathcal{J}), c_I \circ u_b = \gamma_I(b)$$

By defining the arrow $g : B \rightarrow \mathcal{C}(A, C)$ such that $\forall b \in B, g(b) = u_b$, g is the unique arrow from the cone (B, γ) to $(\mathcal{C}(A, C), (\mathcal{C}(A, c_1), \dots, \mathcal{C}(A, c_N)))$, determined by the uniqueness of each u_b .

$$\begin{array}{ccc}
& & \mathcal{C}(A, D I) \\
& \nearrow \gamma_I & \uparrow \mathcal{C}(A, c_I) \\
B & \xrightarrow{g} & \mathcal{C}(A, C)
\end{array}$$

Thus, $(\mathcal{C}(A, C), (\mathcal{C}(A, c_1), \dots, \mathcal{C}(A, c_N)))$ is the finite limit to $\mathcal{C}(A, -) \circ D$.

□

Corollary 1.2. *Since all representable functors are naturally isomorphic to the covariant Hom-functor at A , $\mathcal{C}(A, -) : C \rightarrow \mathbf{Set}$ for some A , representable functors then preserve finite limits.*

Proposition 2.3. *The natural transformation $\Delta : Id \rightarrow F$ given by arrows $\Delta_X : X \rightarrow X \times X := x \mapsto (x, x)$ is the only natural transformation of the type $Id \rightarrow F$.*

$$\begin{array}{ccccc}
\text{Id } A = A & \xrightarrow{\text{Id } f=f} & \text{Id } B = B \\
\downarrow \Delta & & \downarrow \Delta_B \\
F A = A \times A & \xrightarrow{F f=f \times f} & F B = B \times B
\end{array}$$
$$\begin{array}{ccccc}
\text{Id} & \text{Id } \{*\} = \{*\} & \xrightarrow{\text{Id } f=f} & \text{Id } A = A & \\
\downarrow \Gamma & \downarrow \Gamma_{\{*\}} & & \downarrow \Gamma_A & \\
F & F \{*\} := \{*\} \times \{*\} & \xrightarrow{F f=f \times f} & F A = A \times A & \\
& & & & \\
& & & & = \{(*, *)\}
\end{array}$$
$$\forall A \in \mathbf{Ob}(\mathbf{Set}) : \Gamma_A(a) = (f \times f)(*, *) = (f(*), f(*)) = (a, a)$$
☐

Define

$$\text{eval} : \mathcal{D}^c \times \mathcal{C} \rightarrow \mathcal{D}$$

$$\mathcal{D}^c \times \mathcal{C} \xrightarrow{\text{eval}} \mathcal{D}$$

Lemma 3.4. *$eval : \mathcal{D}^c \times \mathcal{C} \rightarrow \mathcal{D}$ is a functor.*

$$\text{eval}(F, -) : \mathcal{C} \rightarrow \mathcal{D}$$

$$\begin{aligned} \text{eval}(F, g \circ f) &= F(g \circ f) \\ &= F\,g \circ F\,f \quad (\text{by functoriality}) \\ &= \text{eval}(F, g) \circ \text{eval}(F, f) \end{aligned} \tag{2}$$

$$\text{eval}(-, C) : \mathcal{D}^c \rightarrow \mathcal{D}$$

$$F C \xrightarrow{\Delta_C} G C \xrightarrow{\Gamma_C} H C$$

$(\Gamma \circ \Delta)_C$

So $\text{eval}(-, C)$ is a functor, and the first part of the lemma is satisfied.

Suppose we have a natural transformation $\Delta : F \rightarrow G$ in $\mathcal{D}^{\mathcal{C}}$, and an arrow $f : C \rightarrow C'$ in $\mathcal{C}$.

$$\begin{array}{ccc} \text{eval}(F, C) & \xrightarrow{\text{eval}(F, f)} & \text{eval}(F, C') \\ \text{eval}(\Delta, C) \downarrow & & \downarrow \text{eval}(\Delta, C') \\ \text{eval}(G, C) & \xrightarrow{\text{eval}(G, f)} & \text{eval}(G, C') \end{array}$$

Then the diagram commutes since the naturality square commutes with Δ being a natural transformation.

$$\begin{array}{ccc} F & C & \xrightarrow{f} F & C' \\ \Delta_C \downarrow & & & \downarrow \Delta_{C'} \\ G & C & \xrightarrow{f} G & C' \end{array}$$

$$\begin{aligned} \text{eval}(F, f) \circ \text{eval}(\Delta, C') &= (F \ f) \circ \Delta'_C \\ &= (G \ f) \circ \Delta_C \quad (\text{by naturality}) \\ &= \text{eval}(G, f) \circ \text{eval}(\Delta, C) \end{aligned} \tag{4}$$

The second part of the lemma is satisfied, and $\text{eval} : \mathcal{D}^{\mathcal{C}} \times \mathcal{C} \rightarrow \mathcal{D}$ is a functor. $\square$

Let $G : \mathcal{E} \times \mathcal{C} \rightarrow \mathcal{D}$ be an arbitrary functor. Define

$$\tilde{G} : \mathcal{E} \rightarrow \mathcal{D}^{\mathcal{C}}$$

$$\tilde{G} :: A \mapsto G(A, -), \quad (f : A \rightarrow B) \mapsto (\Delta : G(A, -) \rightarrow G(B, -))$$

where $G(A, -) : \mathcal{C} \rightarrow \mathcal{D}$ is a functor for each A in $\mathcal{E}$.

Lemma 3.5. *For a given $G : \mathcal{E} \times \mathcal{C} \rightarrow \mathcal{D}$, $\tilde{G} : \mathcal{E} \rightarrow \mathcal{D}^{\mathcal{C}}$ is the unique functor where the following diagram commutes.*

$$\begin{array}{ccc} \mathcal{D}^{\mathcal{C}} \times \mathcal{C} & \xrightarrow{\text{eval}} & \mathcal{D} \\ \tilde{G} \times 1_{\mathcal{C}} \uparrow & \nearrow G & \\ \mathcal{E} \times \mathcal{C} & & \end{array}$$

Proof. Any object A in $\mathcal{E}$ will get mapped to $G(A, -)$ in $\mathcal{D}^{\mathcal{C}}$. Suppose $A \xrightarrow{f} A' \xrightarrow{g} A''$ in $\mathcal{E}$. Then

$$\begin{aligned} \forall C \in \text{Ob}(\mathcal{C}), ((\tilde{G} \ g) \circ (\tilde{G} \ f))(C) &= (\tilde{G} \ g)(C) \circ (\tilde{G} \ f)(C) \\ &= (\Delta'_C : G(A', C) \rightarrow G(A'', C)) \circ (\Delta_C : G(A, C) \rightarrow G(A', C)) \\ &= (\Delta' \circ \Delta)_C : G(A, C) \rightarrow G(A'', C) \\ &= \tilde{G}(g \circ f)(C) \end{aligned} \tag{5}$$

$$\tilde{G} \ 1_A = 1_{G(A, -)} = 1_{\tilde{G} \ A}$$

So $\tilde{G} : \mathcal{E} \rightarrow \mathcal{D}^{\mathcal{C}}$ is a functor.

Let (E, C) be an object in $\mathcal{E} \times \mathcal{C}$.

$$\text{eval} \circ (\tilde{G} \times 1_{\mathcal{C}})(E, C) = \text{eval}(G(E, -), C) = G(E, C)$$

Let (f, g) be an arrow in $\mathcal{E} \times \mathcal{C}$.

$$\begin{aligned} \mathbf{eval} \circ (\tilde{G} \times 1_{\mathcal{C}})(f : A \rightarrow A', g : C \rightarrow C') &= \mathbf{eval}(\Delta : G(A, -) \rightarrow G(A', -), g : C \rightarrow C') \\ &= k : G(A, C) \rightarrow G(A', C') \\ &= G(f : A \rightarrow A', g : C \rightarrow C') \end{aligned} \tag{6}$$

So $\mathbf{eval} \circ (\tilde{G} \times 1_{\mathcal{C}}) = G$.

To show uniqueness, suppose there is some other functor $H : \mathcal{E} \rightarrow \mathcal{D}^{\mathcal{C}}$ such that $\mathbf{eval} \circ (H \times 1_{\mathcal{C}}) = G$. Then for each object (E, C) in $\mathcal{E} \times \mathcal{C}$,

$$\mathbf{eval} \circ (H \times 1_{\mathcal{C}})(E, C) = \mathbf{eval}(H(E), C) = H(E)(C) = G(E, C)$$

So $\forall (E, C) \in \text{Ob}(\mathcal{E} \times \mathcal{C}), H(E)(C) = G(E, C)$. Then for each arrow (f, g) in $\mathcal{E} \times \mathcal{C}$,

$$\begin{aligned} \mathbf{eval} \circ (H \times 1_{\mathcal{C}})(f : A \rightarrow A', g : C \rightarrow C') &= \mathbf{eval}(H f : H A \rightarrow H A', g : C \rightarrow C') \\ &= k : H A C \rightarrow H A' C' \\ &= G(f : A \rightarrow A', g : C \rightarrow C') \\ &= k : G(A, C) \rightarrow G(A', C') \end{aligned} \tag{7}$$

So $\forall (f, g) \in \text{Ob}(\mathcal{E} \times \mathcal{C}), H(f)(g) = G(f, g)$. Thus, $H = \tilde{G}$ is unique. $\square$

Proposition 3.6. *Func($\mathcal{C}, \mathcal{D}$) is an exponential object in **Cat**.*

Proof. By Lemmas 3.4, 3.5, there is an object $\mathcal{D}^{\mathcal{C}}$ and morphism $\mathbf{eval} : \mathcal{D}^{\mathcal{C}} \times \mathcal{C} \rightarrow \mathcal{D}$, where for every $G : \mathcal{E} \times \mathcal{C} \rightarrow \mathcal{D}$, there is a unique morphism $\tilde{G} : \mathcal{E} \rightarrow \mathcal{D}^{\mathcal{C}}$ such that $\mathbf{eval} \circ (\tilde{G} \times 1_{\mathcal{C}}) = G$. $\text{Func}(\mathcal{C}, \mathcal{D})$ is an exponential object in **Cat**. $\square$

Question 4

(a)

Proposition 4.7. $a \Rightarrow b$ is an exponential.

Proof.

$$\mathbf{eval} : (a \Rightarrow b) \wedge a \leq b$$

Let $f : c \wedge a \leq b$ be an arrow. We want to show the unique arrow $\tilde{f} : c \leq (a \Rightarrow b)$ such that

$$\mathbf{eval} \circ (\tilde{f} \times 1_a) = f$$

$$\begin{aligned} \mathbf{eval} \circ (\tilde{f} \times 1_a) &= \mathbf{eval} \circ (c \leq (a \Rightarrow b) \wedge a \leq a) \\ &= ((a \Rightarrow b) \wedge a \leq b) \circ (c \wedge a \leq (a \Rightarrow b) \wedge a) \\ &= c \wedge a \leq b \\ &= f \end{aligned} \tag{8}$$

Since B is a poset, there is at most 1 arrow between objects - $\tilde{f}$ is a unique arrow.
 $a \Rightarrow b$ is an exponential.

□

(b)

$$\perp = \emptyset$$

$$\top = X$$

(c)

The atoms are the elements of the set X as sets. i.e. the set $\{x\}$ where $x \in X$.

(d)

Proposition 4.8. $t : Id \rightarrow At \circ \mathcal{P}^B$ is a natural isomorphism.

$$\begin{array}{ccccc}
 Id & & A & \xrightarrow{\quad f \quad} & B \\
 \downarrow t & & \downarrow t_A & & \downarrow t_B \\
 At \circ \mathcal{P}^B & & \{\{a\} | a \in A\} & \xrightarrow{\{a\} \mapsto \{f(a)\}} & \{\{b\} | b \in B\}
 \end{array}$$

$\begin{array}{ccc} a & \mapsto & f(a) \\ \downarrow & & \downarrow \\ \{a\} & \mapsto & \{f(a)\} \end{array}$

Proof. To prove the naturality square commutes, we show that for all finite sets A, B ,

$$((At \circ \mathcal{P}^B) f) \circ t_A = t_B \circ f$$

$$\mathbf{FinSet} \xrightarrow{\mathcal{P}^B} \mathbf{FinBool}^{op} \xrightarrow{At} \mathbf{FinSet}$$

$$\begin{array}{ccccc}
 A & \xrightarrow{\quad} & \mathcal{P}(A) & \xrightarrow{\quad} & \{\{a\} | a \in A\} \\
 f \downarrow & & \uparrow h(T) = \{a \in A | f(a) \in T\} & & \downarrow At(h) \\
 B & \xrightarrow{\quad} & \mathcal{P}(B) & \xrightarrow{\quad} & \{\{b\} | b \in B\}
 \end{array}$$

Note that by definition of the functors $\mathcal{P}^B$ and At , where $\mathcal{P}^B f = h : \mathcal{P}(B) \rightarrow \mathcal{P}(A)$,

$$\begin{aligned}
 \forall a \in A, (\forall T \in \mathcal{P}(B), f(a) \in T &\iff a \in h(T) \iff \{a\} \subseteq h(T)) \\
 &\iff (\exists! \{b'\} \in \mathcal{P}(B) : \{b'\} \subseteq T \iff b' \in T)
 \end{aligned} \tag{9}$$

In other words, for all elements $a \in A$, let its corresponding atom $\{a\} \subseteq h(T)$ for all subsets $T \subseteq B$ where $f(a) \in T$, so that $\{b'\} \subseteq B$ is the unique atom that satisfies the property of a Boolean algebra homomorphism.

It must be that $b' = f(a)$ since $\{f(a)\} \subseteq B$ is one such atom that satisfies the property, and by uniqueness. So $\forall a \in A, At(h)(\{a\}) = \{f(a)\} \subseteq B$. t is a natural transformation.

To show the natural isomorphism, we can define the inverse for all finite sets A ,

$$t_A^{-1} : \{\{a\} | a \in A\} \rightarrow A$$

$$t_A^{-1} :: \{a\} \mapsto a, \quad (k : \{a\} \mapsto \{b\}) \mapsto (f(a) = b)$$

so $t_A^{-1} \circ t_A = 1_{Id \ A}$, $t_A \circ t_A^{-1} = 1_{At \circ \mathcal{P}^B \ A}$. We have for all A , t_A is an isomorphism, so t is a natural isomorphism. □

Optional

(a)

Suppose we have a function $f : A \rightarrow B$. Then $[f] \circ \text{flatten}_A = \text{flatten}_B \circ (\text{Tree } f)$ since each element in the underlying tree will get mapped to the same element in the resulting list.

(b)

If we have the following where `undef1 . id = undef2`,

```
1 undef1 = undefined :: a -> b
2 undef2 = \_ -> undefined
3 seq undef1 () = undefined
4 seq undef2 () = ()
```

Different values are given with `seq`. The identity axiom is not satisfied so **Hask** is not a category.